**MAJO BRIDGE**
Technologies and Events PLC

PROPOSAL TO THE BOARD OF DIRECTORS

# Technical Proposal

## MaJo e-School Bridge — the Buna Digital School Platform

Bank-sponsored school management, parent engagement and fee payment platform

SUBMITTED TO

**Buna International Bank S.C.**
Board of Directors and Executive Management

SUBMITTED BY

**Majo Bridge Technologies and Events PLC**
Makers of MaJo e-School Bridge

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FOR THE BOARD OF DIRECTORS

# Board summary

Buna International Bank proposes to give schools that bank with it a complete school management platform free of charge. In return, the schools and their parents route fee payments and school finances through Buna. MaJo e-School Bridge is that platform: it is already built, in production, and runs every office of a school on web and mobile in English, Amharic and Afaan Oromo.

| 61 | 6 | 3 | 6 |
|---|---|---|---|
| modules live today | portals: admin, teacher, parent, student, driver, console | languages: English, Amharic, Afaan Oromo | platforms: web, Android, iOS, Windows, macOS, Linux |

### What the Board is asked to note

- **A ready product, not a project.** 61 modules covering academics, students, finance, HR, payroll, transport, communication and compliance are live today.
- **Every parent becomes a Buna touchpoint.** The parent app is used daily for attendance, results, messages and the school bus, so the 'Pay with Buna' button sits where parents already are.
- **Every school transaction can be a Buna transaction.** Student fees, materials, payroll, supplier payments and petty cash are already recorded in the platform's double-entry ledger; the proposed Buna integration makes Buna the default rail for each of them.
- **Multi-tenant by design.** One platform serves many schools; each school's data is separated in the database itself and checked on every request.
- **Bank-grade programme.** An independent penetration test, closure of all open audit findings, alignment with the Personal Data Protection Proclamation No. 1321/2024 and NBE expectations, and an optional Buna-dedicated environment are part of this proposal.
- **Fast to market.** Buna integration and security hardening in 8 weeks, a 5-school pilot, then rollout in waves; a new school is onboarded in about two weeks.

# 01 Introduction and purpose

This proposal describes the technology behind MaJo e-School Bridge, the modules and their strengths, how the platform keeps each school's data separate and secure, and how it will be integrated with Buna International Bank so that school payments flow through the Bank. Prices and commercial terms are in the separate Financial Proposal MBT-BUNA-FIN-2026-01.

## 1.1 About Majo Bridge Technologies and Events PLC

Majo Bridge is an Ethiopian technology company that designs, builds and operates MaJo e-School Bridge. The company owns the full source code, builds and supports the product in Addis Ababa, and deploys it for private and international schools. Because the team owns the code, the Bank deals with one accountable partner for the product, the integration, support and future development.

## 1.2 Scope of this proposal

- **Platform.** The complete MaJo e-School Bridge platform with all modules and portals, hosted and maintained by Majo Bridge.
- **Buna integration.** Payment references, Pay with Buna, payment confirmation, reconciliation, payroll and account-opening referral (Section 8).
- **Bank partner portal.** A Buna view of enrolled schools, adoption and aggregated payment volumes.
- **Rollout.** Pilot, school onboarding, data migration, training and support for every school the Bank introduces.
- **Assurance.** Security hardening, independent testing and the controls set out in Sections 9 and 10.

# 02 The partnership model

The model has three parties. Each one gains something it does not have today.

| Party | Gives | Gets |
|---|---|---|
| Buna International Bank | Sponsors the platform for its school customers; promotes it through relationship managers and branches | Schools, parents and staff as customers; school fee and payroll flows; daily digital engagement with families; data for credit and product decisions (with consent) |
| School | Banks its fee collection and payroll with Buna; asks parents to pay through Buna | A full school management system at no cost; automatic fee reconciliation; less cash handling; better parent communication |
| Parent | Pays school fees and school purchases through Buna channels | Real-time view of attendance, results, homework, fees and the school bus; instant digital receipts; no queues at the school finance office |
| Majo Bridge | Provides, hosts, integrates, supports and develops the platform | A monthly fee per enrolled student, paid by the Bank (see the Financial Proposal) |

## 2.1 How money moves

1. The school sets its fee structure in the Finance module. The platform issues an invoice per student per term or month, each with a unique Buna payment reference.
2. The parent opens the app, sees the invoice and taps Pay with Buna. The parent pays through the Buna mobile app, USSD, internet banking, a Buna branch or a Buna agent, quoting the reference.
3. Buna confirms the payment to the platform in real time through a secure interface. The invoice is marked paid, the parent receives a digital receipt, and the school's ledger is posted automatically.
4. Funds settle to the school's Buna account. Each morning the platform and the Bank exchange a reconciliation file, so the school finance office has nothing to match by hand.
5. At month end the school runs payroll in the platform and sends a bulk salary file to Buna, so staff salaries are paid into Buna accounts.

## 2.2 Why this works for the Bank

| Benefit to Buna | How the platform delivers it |
|---|---|
| New retail customers | Every parent who signs up in the app is invited to open or link a Buna account; account-opening referrals are tracked per school |
| Deposits and float | School fee collections settle into the school's Buna account; payroll keeps staff salaries in Buna; parents keep balances for recurring fees |
| Transaction volume | Tuition, registration, transport, books and materials, uniforms, events, payroll, supplier payments and petty-cash top-ups — all can carry a Buna reference |
| Daily digital engagement | Parents open the app for attendance, results, messages and the bus — not only when fees are due — keeping Buna's brand in front of them every school day |
| Cross-sell | School-fee instalment loans for parents; salary-advance and personal loans for teachers; working-capital and asset finance for schools; savings products for students |

| Benefit to Buna | How the platform delivers it |
|---|---|
| Better credit decisions | With the customer's consent, fee-payment history and school cash flow give the Bank evidence for lending to parents and schools |
| Customer stickiness | A school that runs its academics, HR and finance on a Buna-sponsored platform has strong reasons to keep its banking with Buna |
| Brand and CSR | 'Powered by Buna' on every school's app and report card; a visible contribution to the digital transformation of Ethiopian education |

## 03 Platform overview

| Dimension | MaJo e-School Bridge |
|---|---|
| Modules | 61 modules in 10 groups, plus the role-based dashboard (Section 5 and Annex A) |
| Portals | Administration and staff web portal (ERP) · Teacher app · Parent app · Student app · Driver app · Platform console for Majo Bridge and the Bank's partner team |
| Devices | Web browser, Android, iPhone and iPad, Windows, macOS and Linux — one code base for all |
| Languages | English, Amharic (አማርኛ) and Afaan Oromo |
| Tenancy | Multi-tenant: one platform, many schools, strict data separation per school (Section 7) |
| Branding | Each school's own logo, name and colours on its login and apps; co-branding 'Powered by Buna' available |
| Offline tolerance | Changes are kept on the device and synchronised automatically when the connection returns |
| Quality | About 990 app tests and role-by-role database permission tests run automatically on every change |

## 04 Modules and their strengths

The platform has 61 modules. Each school can switch modules on or off for its own needs. The table summarises each group; Annex A lists every module.

| Group | Modules | Strength for the school | Value for Buna |
|---|---|---|---|
| Organization | 4 | One account for a whole school group, its campuses and cameras | Group-level banking relationship |
| Academics | 16 | Timetable, markbook, exams, approvals, report cards, LMS, lesson plans | Daily parent and teacher usage |
| Student Services | 13 | Admissions to alumni; verified parent–child links; safeguarding; clinic | Verified parent identities linked to each student |

| Group | Modules | Strength for the school | Value for Buna |
|---|---|---|---|
| Finance | 2 | Fees, invoices, receipts, cashboxes, expenses, budgets, payables, full ledger and statements; procurement and store | Fee, supplier and petty-cash flows on the Buna rail |
| HR and Transport | 9 | Staff, leave, contracts, payroll with Ethiopian tax and pension, buses, live GPS | Payroll accounts; transport fees |
| Learning Resources | 2 | Library and paid e-books and materials | Small, frequent purchases |
| Communication | 5 | Announcements, events, gallery, calendar, secure messaging | Channel for Bank campaigns agreed with schools |
| Quality & Insights | 3 | QA plans, analytics, AI assistant | Aggregated insight for the partner portal |
| System | 6 | Role permissions, audit log, health, compliance | Controls the Bank's risk team expects |
| Account | 1 | Profile and two-factor authentication | Secure access |

### 4.1 Strengths that set the platform apart

- **Complete school ERP.** Academics, student services, finance, HR, payroll, transport and communication in one product — schools do not need several systems.
- **Accounting-grade finance.** Every fee, payment, expense and payroll posts to a double-entry ledger with trial balance, income and expenditure, balance sheet and cash-flow statements, so a Buna payment confirmation lands directly in the school's books.
- **Ethiopian context.** Ethiopian payroll income tax and pension, Ethiopian Birr throughout, three local languages, and support from Addis Ababa.
- **Approval workflows.** Grades, expenses, payables, procurement, transfers and parent links all move through approval chains with a full audit trail.
- **Parents who are actually engaged.** Attendance alerts, results, homework, messages, events and the live school bus give parents a reason to open the app every day.
- **Safeguarding and privacy.** Child-protection cases, clinic notes and discipline records are restricted to named roles.
- **Works on any device.** The same platform runs in a browser and as native apps, with tolerance for unstable internet.
- **AI assistance.** The Maya Assistant helps staff find information and draft communication.

## 05 Technical architecture

### 5.1 Frontend

- **Framework.** Flutter (Dart 3) — a single code base compiled to native Android, iOS, Windows, macOS and Linux apps and to a web application.
- **Design.** Role-based dashboards, responsive layouts for phone, tablet and desktop, accessible contrast and font sizes, and per-school branding.

- **Localisation.** All screens available in English, Amharic and Afaan Oromo.
- **Resilience.** Local storage with an outbox; every change is queued on the device and synchronised every few seconds, so work is not lost when the network drops.
- **Distribution.** Google Play, Apple App Store, and web access on a school sub-domain (for example fenote-raey.majobridge.com); a Buna-branded domain can be used.

## 5.2 Backend

- **Database.** PostgreSQL managed through Supabase, with row-level security enforced inside the database for every table.
- **Authentication.** Supabase Auth issuing signed JSON Web Tokens (JWT) that carry the user's school and role.
- **Server functions.** 25 serverless Edge Functions (TypeScript) handle sensitive operations — school login, account creation, password reset, parent invitation and verification, school provisioning, logo upload, push notification, AI assistant and crash reporting. Privileged keys never reach the user's device.
- **Real time.** Live channels for messaging and bus GPS.
- **Files.** Private storage for school documents, served through short-lived signed links; a separate public bucket only for school logos.
- **Notifications.** Firebase Cloud Messaging for push to Android, iOS and web.
- **AI.** Google Gemini through a server function; no school data is used to train external models.

## 5.3 Hosting and operations

| Layer | Current production | Option for Buna |
|---|---|---|
| Web application | Cloudflare Pages global edge network with TLS | Same, on a Buna-branded domain |
| Database and functions | Supabase managed cloud | Buna-dedicated Supabase project, or self-hosted in an Ethiopian data centre or Buna's own data centre |
| Mobile apps | Google Play and Apple App Store | Same; optional Buna co-branded build |
| Bank integration | — | Integration service in Buna's DMZ or Majo Bridge's environment, connected over mutual TLS and IP allow-listing |
| Monitoring | Crash reporting, system health page, CI checks | Plus uptime monitoring, alerting and monthly service reports to the Bank |

## 5.4 Development and release process

- **Source control and continuous integration.** Every change runs static analysis, about 990 automated app tests and database permission tests that check each user role and cross-school access. A further isolation test against a live staging environment is part of the go-live plan for the bank.
- **Load testing.** K6 load-test scripts exercise login, reads and synchronisation at school scale before major releases.

- **Environments.** Separate development, staging and production; database changes are versioned migrations.
- **Change control.** Changes that affect the Bank integration are released only after joint testing and the Bank's approval.

# 06 Multi-tenancy

Yes — MaJo e-School Bridge is a multi-tenant platform. One installation serves many schools, and each school is a separate tenant with its own data, users, settings, branding and modules.

## 6.1 How tenants are separated

1. Every school has a unique School ID (for example FN0597). Every record in the database carries the School ID of the school that owns it.
2. When a user signs in, the server issues a signed token that contains the user's School ID and role. The device cannot change it.
3. PostgreSQL row-level security compares the School ID on each record with the School ID in the token on every read and every write. A user of one school cannot read or change another school's data, even through a modified app — the database itself refuses.
4. Automated database tests act as each user role and as a user of another school, and confirm that cross-school access and actions outside the user's role are refused. They run on every change. The same check against a live staging environment is part of the go-live plan.
5. Files are stored under the school's own folder and served only through signed, time-limited links.

## 6.2 Tenant management

- **Provisioning.** The platform console creates a new school, its administrator account, logo and module selection in minutes.
- **Per-school configuration.** Academic years, grading schemes, fee structures, approval chains, roles and permissions are set per school.
- **School groups.** A group with several schools or campuses can be managed from one institution account while each school's data stays separate.
- **Bank view.** The Buna partner portal sees only aggregated, non-personal figures for schools in the Buna programme (Section 8.4).

### 6.3 Deployment options for the Bank

| Option | Description | When to choose |
|---|---|---|
| A. Shared cloud (default) | Buna schools run on Majo Bridge's managed multi-tenant platform | Fastest launch; lowest cost |
| B. Buna-dedicated environment | A separate database and functions used only by Buna-sponsored schools, branded for Buna | When the Bank wants its programme isolated from other schools |
| C. In-country / on-premises | The platform's backend installed in an Ethiopian data centre or Buna's data centre | When data-residency or NBE requirements call for it |

## 07 Buna International Bank integration

Today schools record bank transfers and mobile-money payments in the Finance module by hand. The Buna integration replaces this with automatic, real-time confirmation from the Bank. It is the main development deliverable of this proposal and is delivered in the first 8 weeks.

### 7.1 Integration services

| Service | What it does | Direction |
|---|---|---|
| Payment reference | A unique, check-digit reference per invoice or purchase, printed on the invoice and shown in the parent app | Platform → parent / Bank |
| Bill presentment | Buna channels (mobile app, USSD, branch, agent) look up the amount due by reference before payment | Bank → platform |
| Pay with Buna | Parent app hands the payment to the Buna mobile app or payment page, then returns to the receipt | Platform ↔ Bank |
| Payment confirmation | Real-time notification of each successful payment; invoice marked paid, receipt issued, ledger posted | Bank → platform |
| Reversal and refund | Reversals and refunds update the invoice and the ledger with a full trail | Bank ↔ platform |
| Daily reconciliation | End-of-day file or interface comparing payments; exceptions shown to the school finance office and the Bank | Bank ↔ platform |
| Settlement report | Daily settlement per school account, matched to the school ledger | Bank → platform |
| Payroll bulk file | Monthly salary file in Buna's format generated from the Payroll module | Platform → Bank |
| Supplier and expense payment | Approved payables and expenses exported for payment from the school's Buna account | Platform → Bank |
| Account-opening referral | Parent and staff consent to be contacted by Buna; referral code tracked per school | Platform → Bank |

### 7.2 Integration security

- Mutual TLS and IP allow-listing between the Bank and the platform.
- Every message signed (HMAC-SHA256 or the Bank's standard) and time-stamped; replayed or altered messages are rejected.
- Idempotency keys so a repeated confirmation never marks a payment twice.
- No card or account credentials ever pass through or are stored in the platform; the parent authenticates only inside Buna's channels.
- Every integration message logged and kept for audit; daily reconciliation closes any gap.

### 7.3 Integration standards

Majo Bridge will adapt to the Bank's existing interface standard (REST/JSON, ISO 20022 messages, or file exchange over SFTP) and to the Bank's core-banking and mobile-banking platforms. Interface specifications are agreed jointly in the first two weeks.

### 7.4 Buna partner portal

- Schools in the programme, onboarding status and active users.
- Billable students per school (the basis of the monthly invoice to the Bank).
- Aggregated payment volume and value through Buna, by school and by channel.
- Parent account-opening referrals and conversions.
- No individual student records, grades, health or safeguarding information is visible to the Bank.

## 08 Security architecture

| Control area | What is in place today |
|---|---|
| Identity | Individual accounts per user; passwords hashed with bcrypt (cost 12); minimum length 10 characters; temporary passwords must be changed at first login |
| Strong authentication | Time-based one-time-password (TOTP) two-factor authentication with recovery codes; additional PIN, stored with PBKDF2, for the platform console |
| Abuse protection | Rate limits on login and password reset; generic error messages that do not reveal whether an account exists |
| Authorisation | Role and module access matrix per school; row-level security in the database for every table |
| Data in transit | TLS on every connection — web, apps, database, storage and functions |
| Data at rest | Encrypted storage at the hosting provider; private file storage with signed, time-limited links |
| Secrets | Service keys held only in server functions, never in the apps |
| Audit | School audit log and platform audit log of sensitive actions with user, time and detail |

| Control area | What is in place today |
|---|---|
| Sensitive records | Safeguarding, clinic and discipline records limited to named roles |
| CCTV | Video stays on the school network; the platform stores only camera details, never footage |
| Secure development | Code review, automated tests and database permission tests on every change; internal security audit maintained |

### 8.1 Bank-grade hardening programme

Before the first Buna school goes live, Majo Bridge will complete the following. Progress is reported to the joint steering committee.

| Item | Deliverable | Timing |
|---|---|---|
| Close audit findings | Every open finding in Majo Bridge's internal security audit closed or formally risk-accepted by both parties | Weeks 1–6 |
| Independent penetration test | Test of the web, mobile apps, functions and Buna interface by a firm acceptable to the Bank; critical and high findings fixed and retested | Weeks 6–8 |
| Bank security review | Buna's information-security team reviews the architecture, the test report and these controls | Week 8 |
| Two-factor authentication | Mandatory for school administrators, finance officers and platform staff | Before pilot |
| Backup and recovery | Daily backups with point-in-time recovery; recovery point objective 24 hours or better; recovery time objective 8 hours; restore drill twice a year | Before pilot, then ongoing |
| Incident response | Written plan; notification of the Bank within 24 hours of a confirmed incident affecting Buna schools | Before pilot |
| Annual assurance | Yearly penetration test and security review | Every year |

## 09 Data protection, ownership and compliance

- **Ownership.** Each school owns its data. Majo Bridge processes it on the school's behalf and only to provide the service.
- **Bank access.** The Bank receives aggregated programme figures and the payment information for its own transactions. Personal data of parents and staff is shared with the Bank only with their recorded consent (for example, for account opening).
- **Children's data.** Student records are never used for marketing. Grades, health, discipline and safeguarding information are never shared with the Bank.
- **Law and regulation.** The programme will follow the Personal Data Protection Proclamation No. 1321/2024 and the directives of the National Bank of Ethiopia that apply to the Bank's outsourcing and digital channels. Majo Bridge will sign the Bank's data-processing and confidentiality agreements.
- **Data residency.** Deployment option C (Section 7.3) keeps all data inside Ethiopia if the Bank requires it.

- **Exit.** A school, or the Bank for the programme, can receive a complete export of its data in standard formats at any time and on termination.

## 10 Service levels and support

| Item | Commitment |
|---|---|
| Availability | 99.5% per month for the platform, excluding announced maintenance outside school hours |
| Support hours | Monday to Saturday, 8:00–18:00 East Africa Time; critical incidents 24 × 7 |
| Channels | Phone, Telegram/WhatsApp, e-mail and in-app; a named account manager for the Bank |
| Response — critical (platform or payments down) | 30 minutes; workaround within 4 hours |
| Response — high (one module or one school affected) | 2 hours; fix within 1 working day |
| Response — normal (question or minor defect) | 1 working day; fix in the next release |
| Reporting | Monthly service report to the Bank: uptime, incidents, tickets, adoption, payment volumes |
| Updates | Regular feature and security updates at no extra cost; changes affecting the Bank interface agreed in advance |

## 11 Implementation plan

| Phase | Weeks | Main activities | Exit criteria |
|---|---|---|---|
| 0. Agreement | — | Partnership agreement, data-processing agreement, steering committee named | Signed agreements |
| 1. Integration and hardening | 1–8 | Interface specification; build of payment reference, confirmation, reconciliation, payroll file and partner portal; security programme; joint testing in the Bank's test environment | Joint test passed; penetration test cleared; Bank security sign-off |
| 2. Pilot | 9–16 | Five schools chosen by the Bank onboarded, data migrated, staff and parents trained, live payments | Pilot report: adoption, payments through Buna, issues closed |
| 3. Rollout wave 1 | 17–28 | Schools introduced by Buna relationship managers onboarded in batches | Agreed number of schools live |
| 4. Scale | 29 onward | Continuous onboarding; quarterly product roadmap reviewed with the Bank | KPIs in Section 12 |

### 11.1 Onboarding one school (about two weeks)

1. Kick-off with the school leadership and the Buna relationship manager.
2. School created in the platform; branding, academic year, grades, subjects and fee structure configured.
3. Students, parents, staff and opening balances imported from the school's spreadsheets.
4. Training for administrators, finance officers, teachers and the school's IT focal person; short guides and videos for parents.
5. Parent invitation campaign with Buna: download the app, link children, pay the first invoice with Buna.
6. Go-live, then two weeks of close support.

## 12 Governance and key performance indicators

A joint steering committee of the Bank's digital banking, retail, IT, risk and compliance leads and Majo Bridge's management meets monthly during the first year, then quarterly.

| KPI | Measured by |
|---|---|
| Schools live in the programme | Partner portal |
| Share of school fee value paid through Buna | Reconciliation reports |
| Parents active in the app each month | Platform analytics |
| New Buna accounts referred by the platform | Referral tracking and Bank records |
| Staff paid through Buna payroll | Payroll files |
| Platform availability and incident response | Monthly service report |

## 13 Risks and mitigation

| Risk | Mitigation |
|---|---|
| Schools slow to adopt | Bank relationship managers and Majo Bridge onboarding team work together; pilot results used as proof; training and parent campaigns |
| Parents keep paying in cash | Pay with Buna inside the app, instant receipts, reminders, and optional Bank incentives for digital payment |
| Integration delays | Interface agreed in week 2; manual recording remains available as a fallback, so schools are never blocked |
| Security or privacy incident | Hardening programme, independent testing, two-factor authentication, incident plan with 24-hour notification |
| Internet instability at schools | Offline-tolerant apps with automatic synchronisation |
| Dependence on one supplier | Source-code escrow option, full data export, documented operations handbook |

## 14 Why Majo Bridge

- A complete, working product already in use by schools — the Bank is not funding a development project.
- One local team that owns the code, so integration and changes are fast.
- A finance module built on accounting principles, ready to receive bank payment confirmations.
- Multi-tenant architecture with database-level isolation, checked by automated database tests on every change.
- Commitment to the Bank's security, compliance and service standards set out in this proposal.

ANNEX

# Annex A — Full module list

| # | Group | Module | What it does |
|---|---|---|---|
| 1 | Organization | Institution Management | Group-level view of every school, campus and head office in one tenant. |
| 2 | Organization | School Management | School profile, academic years, terms, grading schemes and branding. |
| 3 | Organization | Campus Management | Several campuses under one school with their own classes and staff. |
| 4 | Organization | CCTV | Register any number of school cameras; footage stays on the school network. |
| 5 | Academics | Academic Management | Grades, sections, subjects and teacher allocations. |
| 6 | Academics | Examinations & Grade Approvals | Exam schedules with head-of-department and principal approval. |
| 7 | Academics | Weighted Markbook | Continuous assessment with weights per task and automatic totals. |
| 8 | Academics | Report Cards | Branded report cards generated from approved marks. |
| 9 | Academics | Exam bank & papers | Question bank and exam paper preparation. |
| 10 | Academics | Attendance | Daily and per-period attendance with parent notification. |
| 11 | Academics | Homework | Assignments, submissions and marking. |
| 12 | Academics | At-risk & insights | Early warning on attendance and results falling behind. |
| 13 | Academics | Analytics & exports | Class and subject analytics with spreadsheet export. |
| 14 | Academics | Classroom Teachers | Teacher directory and workload. |
| 15 | Academics | Add Teacher | Guided teacher onboarding with secure first login. |
| 16 | Academics | Timetable | Class and teacher timetables without clashes. |
| 17 | Academics | LMS | Online courses, lessons, quizzes and learning materials. |
| 18 | Academics | Lesson plans | Plans submitted, reviewed and shared with students. |
| 19 | Academics | Curriculum office | Curriculum coverage and quality review. |
| 20 | Academics | Grade Workflow | Marks move from teacher to approval to publication with a full trail. |
| 21 | Student Services | Students | Student information system with documents and history. |
| 22 | Student Services | Admissions | Online application, review and enrolment. |
| 23 | Student Services | Alumni | Graduates kept in touch with the school. |

| # | Group | Module | What it does |
|---|---|---|---|
| 24 | Student Services | Add Student | Fast single or bulk enrolment. |
| 25 | Student Services | Student Portal Settings | Controls what students see and do in their app. |
| 26 | Student Services | Student Password Resets | Secure, audited password resets for students. |
| 27 | Student Services | Student Affairs & Discipline | Incidents, actions and parent communication. |
| 28 | Student Services | Student support | Counselling and support plans. |
| 29 | Student Services | Clinic reports | School clinic visits and health notes. |
| 30 | Student Services | Safeguarding | Confidential child-protection cases with restricted access. |
| 31 | Student Services | Student programs | Clubs, sports and special programmes. |
| 32 | Student Services | Parents & Link Approvals | Parents verified and linked to their children before they see any data. |
| 33 | Student Services | Transfers & Promotion | End-of-year promotion and transfers with approval. |
| 34 | Finance | Finance | Fee structures, student invoices, payments with receipt numbers, cashboxes, expenses with approval, budgets, payables, double-entry ledger, trial balance, income and expenditure, balance sheet, cash flow, and parent fee statements. |
| 35 | Finance | Inventory, Procurement & Store | Purchase requests, approvals, suppliers, store stock and issues. |
| 36 | Human Resources and Transport | Human Resource | Staff records, positions and documents. |
| 37 | Human Resources and Transport | Leave & contracts | Leave requests, approvals and contract tracking. |
| 38 | Human Resources and Transport | Payroll | Monthly payroll with Ethiopian income tax and pension, payslips and Excel/PDF export. |
| 39 | Human Resources and Transport | Administration Staff Directory | Non-teaching staff with their roles. |
| 40 | Human Resources and Transport | Add Administration Staff | Guided onboarding with role-based access. |
| 41 | Human Resources and Transport | Transport | Routes, stops and student assignments. |
| 42 | Human Resources and Transport | Buses | Fleet register and capacity. |
| 43 | Human Resources and Transport | Live GPS | Parents see the school bus on a live map. |
| 44 | Human Resources and Transport | Register Driver | Driver accounts for the driver app. |
| 45 | Learning Resources | Library | Catalogue, loans and returns. |

| # | Group | Module | What it does |
|---|---|---|---|
| 46 | Learning Resources | e-Books & Materials | Digital books and paid learning materials. |
| 47 | Communication | Announcements | School-wide or class-specific notices with push notification. |
| 48 | Communication | Events | School events with invitations. |
| 49 | Communication | Gallery | Photos shared with families. |
| 50 | Communication | Calendar | Academic and event calendar. |
| 51 | Communication | Messages | Secure teacher–parent messaging in real time. |
| 52 | Quality & Insights | QA Findings & Plans | Quality-assurance findings and improvement plans. |
| 53 | Quality & Insights | Reports & Analytics | School-wide dashboards and reports. |
| 54 | Quality & Insights | Maya Assistant | AI assistant that answers questions and drafts content for staff. |
| 55 | System | Role Permissions | Who can see and do what, per role and per module. |
| 56 | System | Audit Log | Every sensitive action recorded with user and time. |
| 57 | System | System Health | Sync, storage and service status. |
| 58 | System | Digital operations | Operational tools for the school's IT officer. |
| 59 | System | Go-live & compliance | Go-live checklist and compliance tracking. |
| 60 | System | Settings | School-level configuration. |
| 61 | Account | Profile | Own profile, password and two-factor authentication. |

ANNEX

# Annex B — Glossary

| Term | Meaning |
|---|---|
| Tenant | One school (or school group) with its own separated data inside the shared platform |
| School ID | Unique code that identifies a school's data, for example FN0597 |
| Row-level security (RLS) | Database rule that checks, for every record, whether the signed-in user may see or change it |
| JWT | Signed token issued at sign-in that carries the user's school and role |
| Edge Function | Small server program that performs a sensitive operation on the server instead of on the device |

| Term | Meaning |
|---|---|
| Reconciliation | Daily matching of payments recorded by the Bank with payments recorded in the platform |
| TOTP | Six-digit code from an authenticator app used as a second sign-in factor |
| RPO / RTO | Recovery point objective (maximum data loss) and recovery time objective (maximum downtime) after a failure |

SUBMISSION

# Sign-off

Submitted on behalf of Majo Bridge Technologies and Events PLC for the consideration of the Board of Directors of Buna International Bank. This Technical Proposal is valid until 2 February 2027.

| For Majo Bridge Technologies and Events PLC | Received for Buna International Bank S.C. |
|---|---|
| Name | Name |
| Title<br>General Manager | Title |
| Signature | Signature |
| Date | Date |
| Company seal | Bank seal |